We aim for a maximum dough temperature of 25°C / 78°F when it comes off the mixer (it should be within 3-4°C / 5-7°F of this temperature). If it is warmer than this temperature, you can place it in the refrigerator or on a marble surface to cool.

DESIRED DOUGH TEMPERATURE

Controlling temperature is one of the prerequisites for great baking, and temperature affects the outcome at several stages of the pizza-making process. It obviously matters a great deal during baking, and also during fermentation and proofing. Some baking books and experts go on at great length about the importance of temperature during mixing as well. They give rules for determining the desired dough temperature (DDT), a target temperature that will let you know when mixing is done. And they offer formulas for calculating how warm or cold the water should be when you add it to the flour—though these formulas usually use an imprecise, empirically derived friction factor that isn't very accurate. But does temperature really matter that much during mixing? We did a set of experiments to find out.

We prepared doughs and varied the mixer used, mix time, and water temperature to see the effects on DDT. Our conclusion is that hitting the DDT on the nose isn't important for most pizza makers in homes, restaurants, and small commercial operations, especially if you can extend your bulk fermentation time. It's much more important to control the temperature of the fermentation environment. We do want to note that pizza dough doesn't immediately cool down in the refrigerator when you cold-proof it. If you start with a dough that is too warm, it may overproof by the time the 48 hours is up. The ideal temperature of the final dough should be 24–25°C / 75–78°F. You should also make sure that all of the ingredients are at around 21°C / 70°F. Otherwise, you might need to start with colder water.

WHAT IS AUTOLYSE?

Autolyse is a two-stage mixing technique that gained currency—and a fancy name—back in 1974, when French baker Raymond Calvel published a technical article that described the method and touted it as a way to deal with extremely strong wheat flours. Calvel coined the term "autolyse" (sometimes spelled "autolyze") from "autolysis," a term used in biology to describe the process of self-digestion that cells undergo as they die and are dismantled by the enzymes inside them. A peach ripening in a fruit basket, for example, softens mainly due to the enzymatic destruction of autolysis. Flour contains natural enzymes, too, and Calvel's idea was that giving those enzymes water and time would boost their activity and allow them to soften up the starches and proteins in the dough.

Definitions of the term and opinions on its use have taken off in many directions in the decades since. This method is most often used in bread bakeries rather than pizzerias. There's considerable disagreement as what to include in an autolyse and whether it's actually useful in a contemporary setting. The more popular variants differ in what initially goes into the flour and water and how long the wet flour is left to sit before salt and other ingredients are added and mixing begins in earnest. Many books even refer to autolyse as a resting period after combining all ingredients. And modern cookbooks from many leading bakers do not include an autolyse step.

There are plenty of ways to mix a proper dough and make a fine baked pizza, and including an autolyse stage is just one of them. Is it worth it? Are there

We perform an autolyse on both our focaccia (top) and high-hydration al taglio (bottom) doughs to give the flour additional time to hydrate. These doughs have the two highest hydration levels in the book.

certain applications where it's best? The whole point of autolyse is to hydrate the flour, which kick-starts enzymatic activity and gluten development, thus reducing mixing time (though perhaps not overall time). Autolyse also reduces the chance of overmixing, which can harm the flavor, crumb texture, and color of the pizza.

That's the argument, anyway. We conducted a series of experiments when we wrote *Modernist Bread* and concluded there is some value to this step for making bread, but it depends on your time, workflow, and what kind of bread you're making. We saw the best results using a methodology that's somewhat different from that developed by Raymond Calvel. In our experiments with French lean breads, autolyse improved volume and produced a dough that was easier to handle, easier to shape, and more attractive, with a pleasing creamy-yellow tint. These benefits exist, but they're also not particularly pronounced. And when we tested this method for pizza, there was practically no benefit for doughs mixed to full gluten development—which is most of our thin- and medium-crust doughs. We did find some benefits of autolyse for high-hydration doughs, like Focaccia Dough (see page 148) and High-Hydration al Taglio Pizza Dough (see page 158). The bottom line is, autolyse isn't anything magical. It was simply Calvel's technique-focused recommendation for boosting volume, but as we've found, there are many other techniques and ingredients that can achieve the same goal (see page 1:324).

THE ORDER OF THE INGREDIENTS

In theory, for simple doughs—such as our Brazilian thin-crust or deep-dish—the order in which you mix the ingredients shouldn't matter. Mindful of time constraints, many pizzaioli learn quickly that it doesn't matter whether the water or flour goes into the mixer first so long as they turn the machine on at low speed (or start mixing by hand) soon after the ingredients are combined.

Some pros recommend starting with the liquid components (water/preferments), adding the yeast next so that it blooms, and then adding the flour last (in some recipes, salt and any oil is added once the dough has mixed for a few minutes). The thinking is that this sequence results in a more efficient mix, and we do use this method in the bulk of our recipes. In our tests, however, the order in which the ingredients were combined made little difference in terms of the final product. For consistency, all our recipes specify adding the ingredients in a particular order, because this forms a habit of following a process, which will come in handy so you don't forget to add ingredients (salt being the one a lot of people forget, followed by yeast). This approach also helps those who are new to mixing and baking develop consistent skills.

There are a few caveats, however. If you're using the autolyse method, mix the salt in some of the water (about 5%–10%) that is called for in the recipe. Mix the flour and the rest of the water (you may bloom commercial yeast in the water or mix it in with the flour to disperse it first) to a shaggy mass, let the dough rest 20–30 minutes, and then add the salt water; doing so will help disperse the salt more evenly throughout the dough.

For nearly all preferments, you'll pour the water into the bowl or tub the dough is being mixed in, then pour the preferment on top of the water. We like to pour our preferments on *top* of the water to see if they float; this shows if the preferment is healthy and ripe. You can dissolve the preferment into the water with your hands by squeezing and stirring it, before adding the flour on top, which may save some mixing later.

For more involved doughs, the order of the ingredients becomes less flexible. Solid ingredients (inclusions) such as nuts and fruit should be added to the dough when it has reached medium gluten development. The inclusions are placed on top of the dough in the mixer, which should be turned to low speed. The dough should continue to mix slowly and only until all the inclusions have been fully incorporated. Inclusions with sharp edges, such as slivered almonds, can potentially tear the dough and cause structural damage.

We recommend withholding the salt from the dough if you are using the autolyse method. Just be sure not to forget to add it in when the autolyse step is complete.